

Vectors and the Geometry of Space

Historical Note

Hermann Grassmann published his *Ausdehnungslehre*, the theory of extension, in 1844, and almost nobody read it. It was abstract, it was philosophical, and it was thirty years ahead of its audience; Grassmann eventually gave up mathematics and became a distinguished Sanskrit scholar instead. The practical vector algebra used by physicists was distilled later, in the 1880s, by Josiah Willard Gibbs and Oliver Heaviside, who wanted a compact way to write Maxwell's equations. What you are about to learn is essentially Gibbs's notation carrying Grassmann's idea.

Chapter 1 claimed that linear algebra studies maps preserving addition and scaling. That claim is empty until we say what objects are being added and scaled. This chapter supplies them.

2.1 Three answers to “what is a vector?”

Ask three people and you will get three answers, all correct.

The physicist's answer. A vector is an arrow: a quantity with magnitude and direction. A displacement of three metres east, a force of ten newtons downward.

The programmer's answer. A vector is a list of numbers, $(3, 1)$ or $(0.2, -1.7, 4.0, \dots)$, of some fixed length.

The mathematician's answer. A vector is an element of a vector space: any object at all that you can add to another of its kind, and scale by a number, subject to a few rules.

The first is where intuition lives, the second is what a computer stores, and the third is what makes the subject powerful, because it lets polynomials, functions and matrices be vectors too. We will take the second as our working definition for now and earn the third in Chapter 2 onwards.

Definition 2.1 (Vectors in $\mathbb{R}^n$). For a positive integer n , the set $\mathbb{R}^n$ consists of all ordered lists of n real numbers,

$$x = (x_1, x_2, \dots, x_n), \quad x_i \in \mathbb{R}.$$

The numbers x_i are the components of x . Two vectors are equal exactly when all their components agree. We write vectors as columns when they meet matrices, and as rows when we are short of space; they are the same object.

Key Idea

A vector is not a point, though we draw it as an arrow to one. A vector is a *displacement*: it has a length and a direction but no fixed home. The arrow from $(1, 1)$ to $(4, 2)$ and the arrow from $(0, 0)$ to $(3, 1)$ are the same vector $(3, 1)$. This is why we may always slide an arrow back to the origin, and why the picture and the list of numbers agree.

2.2 Adding and scaling

Everything in this book is built from two operations, so we state them carefully.

Definition 2.2 (Addition and scalar multiplication). For $x, y \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$,

$$x + y = (x_1 + y_1, \dots, x_n + y_n), \quad \lambda x = (\lambda x_1, \dots, \lambda x_n).$$

Both act componentwise. We write $-x$ for $(-1)x$ and $x - y$ for $x + (-y)$.

Geometrically, addition is the instruction *do one displacement, then the other*. Place the tail of y at the tip of x ; the sum runs from the start of the first to the end of the second. Equivalently, draw both from the origin and complete the parallelogram: the diagonal is $x + y$. The two pictures agree because the opposite sides of a parallelogram are the same displacement. Figure 2.1 shows both constructions on the same pair of vectors, and the same point at the end of each.

Example 2.3 (Arithmetic in $\mathbb{R}^2$). Let $u = (3, 1)$ and $v = (-1, 2)$. Then

$$u + v = (2, 3), \quad u - v = (4, -1), \quad 2u = (6, 2), \quad 2u - v = (7, 0).$$

Notice the last one: scaling u by 2 doubles its length and keeps its direction, and subtracting

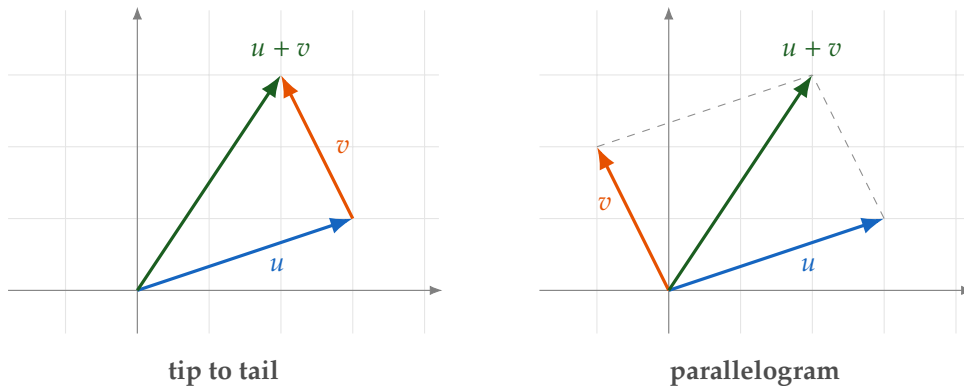


Figure 2.1. Two pictures of the same sum, with $u = (3, 1)$ and $v = (-1, 2)$, so $u + v = (2, 3)$. On the left, v is carried to the tip of u ; on the right, both start at the origin and the sum is the diagonal. The dashed sides are copies of u and v , which is exactly why the pictures agree.

v then lands us exactly on the horizontal axis. Nothing here is deep, but the habit of checking a computation against the picture is worth forming now.

Example 2.4 (Vectors that are not arrows). *The definition asks only for componentwise addition and scaling, so many things qualify.*

- (i) *A greyscale image with 28×28 pixels is a vector in $\mathbb{R}^{784}$: adding two images brightens where both are bright, and scaling by $\frac{1}{2}$ halves the brightness.*
- (ii) *A polynomial $a_0 + a_1t + a_2t^2$ is a vector, since polynomials add and scale.*
- (iii) *A word embedding in a language model is a vector in $\mathbb{R}^{768}$ or larger, and the famous observation that $\text{king} - \text{man} + \text{woman}$ lands near queen is nothing but Definition 2.2 applied to those lists.*

None of these is an arrow in space, and all of them obey the same algebra. That is the whole argument for abstraction.

2.3 Linear combinations and span

We now meet the single most important construction in the subject.

Definition 2.5 (Linear combination and span). *A linear combination of $v_1, \dots, v_k \in$*

$\mathbb{R}^n$ is any vector of the form

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_k v_k, \quad \lambda_i \in \mathbb{R}.$$

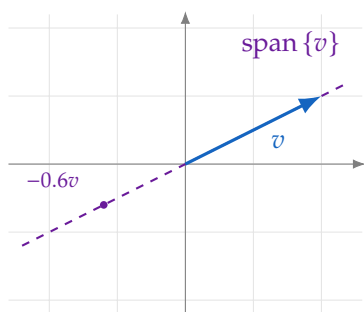
The set of all such combinations is called the span:

$$\text{span}\{v_1, \dots, v_k\} = \left\{ \sum_{i=1}^k \lambda_i v_i : \lambda_i \in \mathbb{R} \right\}.$$

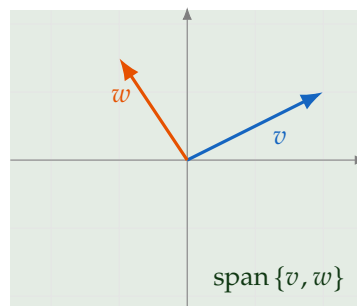
Key Idea

Span answers the question *where can I get to?* Given some vectors and permission to scale and add them, the span is the set of destinations you can reach. Almost every question in the first half of this book is a span question in disguise: solving $Ax = b$ asks whether b lies in the span of the columns of A .

The span of a single nonzero vector is the line through the origin in its direction: scaling stretches, reverses and shrinks it, but cannot turn it. Add a second vector pointing a genuinely different way and the reachable set jumps up a dimension. Figure 2.2 draws both cases side by side, one vector reaching a line and two reaching the whole plane.



one vector: a line



two directions: the plane

Figure 2.2. Span is the set of reachable points. One nonzero vector reaches a line; two vectors pointing in genuinely different directions reach the whole plane, shaded here. Had w been a multiple of v , the second picture would have collapsed back to the first.

Example 2.6 (Is a vector in the span?). Is $b = (7, 0)$ in $\text{span}\{(3, 1), (-1, 2)\}$? We

need λ, μ with

$$\lambda(3, 1) + \mu(-1, 2) = (7, 0) \iff \begin{cases} 3\lambda - \mu = 7, \\ \lambda + 2\mu = 0. \end{cases}$$

The second equation gives $\lambda = -2\mu$; substituting, $-6\mu - \mu = 7$, so $\mu = -1$ and $\lambda = 2$. Indeed $2(3, 1) - (-1, 2) = (7, 0)$, which is the computation we already did in Example 2.3. So yes, and the coefficients are unique.

Example 2.7 (When the span collapses). Consider $\text{span}\{(1, 2), (2, 4)\}$. The second vector is twice the first, so every combination $\lambda(1, 2) + \mu(2, 4) = (\lambda + 2\mu)(1, 2)$ is again a multiple of $(1, 2)$. Two vectors, but only a line: the second one contributed nothing. Detecting this collapse is the subject of linear independence, which we take up in Chapter 6.

2.4 Length, and the dot product

So far we have used only addition and scaling, and with those alone we cannot speak of length or angle. One extra operation supplies both.

Definition 2.8 (Dot product and norm). For $x, y \in \mathbb{R}^n$, the dot product and the norm are

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i, \quad \|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x_1^2 + \cdots + x_n^2}.$$

The dot product is symmetric, is linear in each argument separately, and satisfies $\langle x, x \rangle \geq 0$ with equality only for $x = 0$.

For $n = 2$ the norm is Pythagoras' theorem and nothing more: the vector $(3, 4)$ has length $\sqrt{9 + 16} = 5$. The dot product is less obviously geometric, so the following identity is the one to remember.

Theorem 2.9 (The dot product measures angle). For nonzero $x, y \in \mathbb{R}^n$,

$$\langle x, y \rangle = \|x\| \|y\| \cos \theta,$$

where $\theta \in [0, \pi]$ is the angle between x and y . In particular $\langle x, y \rangle = 0$ exactly when the two are perpendicular.

Proof. Apply the law of cosines to the triangle with sides x , y and $x - y$:

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\|\cos\theta.$$

Expanding the left side with the definition, $\|x - y\|^2 = \langle x - y, x - y \rangle = \|x\|^2 - 2\langle x, y \rangle + \|y\|^2$. Comparing the two expressions, the $\|x\|^2$ and $\|y\|^2$ cancel, leaving $-2\langle x, y \rangle = -2\|x\|\|y\|\cos\theta$, which is the claim. $\square$

Example 2.10 (An angle, computed). Take $u = (3, 1)$ and $v = (-1, 2)$ again. Then

$$\langle u, v \rangle = 3(-1) + 1(2) = -1, \quad \|u\| = \sqrt{10}, \quad \|v\| = \sqrt{5},$$

so $\cos\theta = -1/\sqrt{50} \approx -0.1414$ and $\theta \approx 98.1^\circ$. The sign of the dot product already told us the answer would exceed a right angle, before any arccosine: a negative dot product means the vectors disagree about direction.

Example 2.11 (Orthogonality in $\mathbb{R}^3$). Let $x = (1, 2, 2)$ and $y = (2, -2, 1)$. Then $\langle x, y \rangle = 2 - 4 + 2 = 0$, so they are perpendicular, and $\|x\| = \sqrt{1 + 4 + 4} = 3$, $\|y\| = \sqrt{4 + 4 + 1} = 3$. Two perpendicular vectors of length 3: a small piece of a rotated coordinate frame, which is exactly how such triples will be used when we reach orthogonal matrices.

Insight

The sign of $\langle x, y \rangle$ is a similarity score, and this is why the dot product is everywhere in machine learning. Positive means the two point broadly the same way, zero means they are unrelated in direction, negative means they oppose. Attention layers score a query against a key with a dot product for precisely this reason; so does every cosine-similarity search over embeddings.

Theorem 2.12 (Cauchy–Schwarz and the triangle inequality). For all $x, y \in \mathbb{R}^n$,

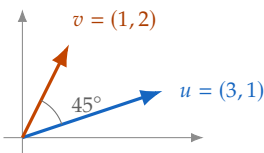
$$|\langle x, y \rangle| \leq \|x\| \|y\|, \quad \text{and consequently} \quad \|x + y\| \leq \|x\| + \|y\|.$$

Proof. If $x = 0$ or $y = 0$ then both sides of the first inequality are zero and there is nothing to prove, so suppose neither is. Then it follows from Theorem 2.9, since $|\cos\theta| \leq 1$. For the second, expand and then apply the first:

$$\|x + y\|^2 = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2. \quad \square$$

The triangle inequality says a detour is never shorter than going direct, which is reassuring, and which will later be the property we demand of any notion of distance.

Figure 2.3 carries out the whole measurement on one pair of vectors: two coordinates each, one sum of products, and an angle of exactly 45 degrees, with no trigonometry used anywhere.



$$\langle u, v \rangle = 5, \|u\| \|v\| = \sqrt{50}, \text{ so } \cos \theta = 5/\sqrt{50} = 1/\sqrt{2}$$

Figure 2.3. The dot product measures an angle, and it does so without any trigonometry. For $u = (3, 1)$ and $v = (1, 2)$ the inner product is 5 and the lengths are $\sqrt{10}$ and $\sqrt{5}$, so $\cos \theta = 5/\sqrt{50} = 1/\sqrt{2}$ and the angle is exactly 45 degrees. The definition only makes sense because $|\langle u, v \rangle| \leq \|u\| \|v\|$ always holds, which is the Cauchy and Schwarz inequality: here $5 \leq 7.071$, with equality only when the vectors are parallel. That inequality is what lets a ratio of an inner product to two lengths be read as a cosine, and it is the reason the notion of angle survives into every inner product space in this book, including spaces of functions where no picture is available.

2.5 Projection: the best approximation on a line

Given a vector b and a direction a , what multiple of a comes closest to b ? This small question is the seed of least squares, of QR factorisation, and of the whole theory of orthogonality in Part V.

Definition 2.13 (Orthogonal projection onto a line). *For $a \neq 0$, the projection of b onto the line $\text{span}\{a\}$ is*

$$\text{proj}_a(b) = \frac{\langle b, a \rangle}{\langle a, a \rangle} a.$$

The coefficient is forced, not chosen. Write the candidate as λa and demand that the error $b - \lambda a$ be perpendicular to a :

$$\langle b - \lambda a, a \rangle = 0 \implies \langle b, a \rangle = \lambda \langle a, a \rangle \implies \lambda = \frac{\langle b, a \rangle}{\langle a, a \rangle}.$$

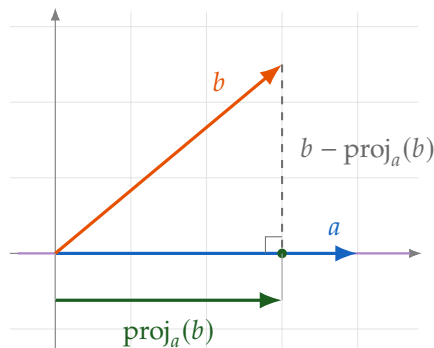


Figure 2.4. The projection of b onto the line through a . The coefficient is whatever makes the dashed error vector perpendicular to a ; that choice, and only that choice, minimises the length of the error.

Example 2.14 (A projection, computed twice). Let $a = (4, 0)$ and $b = (3, 2.5)$, as drawn in Figure 2.4. Then $\langle b, a \rangle = 12$ and $\langle a, a \rangle = 16$, so

$$\text{proj}_a(b) = \frac{12}{16} (4, 0) = (3, 0),$$

and the error is $b - (3, 0) = (0, 2.5)$, which is indeed perpendicular to a . For a less convenient direction take $a = (1, 1)$ and $b = (3, 4)$: now $\langle b, a \rangle = 7$, $\langle a, a \rangle = 2$, and $\text{proj}_a(b) = \frac{7}{2}(1, 1) = (3.5, 3.5)$, with error $(-0.5, 0.5)$. Check: $\langle (-0.5, 0.5), (1, 1) \rangle = 0$, as required.

Caution

The denominator $\langle a, a \rangle$ is not decoration. If a is not a unit vector, forgetting to divide scales your answer by $\|a\|^2$, and it is one of the most common slips in the subject. It disappears only when $\|a\| = 1$, which is exactly why we will work with orthonormal bases whenever we can.

2.6 Lines and planes

With linear combinations in hand, the basic objects of geometry become easy to describe.

A line through the point p with direction $d \neq 0$ is the set $\{p + td : t \in \mathbb{R}\}$. When $p = 0$ this is exactly $\text{span}\{d\}$; otherwise it is that line slid off the origin. For instance the line through $(1, 2)$ with direction $(2, -1)$ consists of the points $(1 + 2t, 2 - t)$, which as t runs over $\mathbb{R}$ traces the whole line.

A plane in $\mathbb{R}^3$ is most compactly described not by what it contains but by what it

is perpendicular to. If n is a nonzero vector and p a point, the plane through p with normal n is

$$\{x \in \mathbb{R}^3 : \langle x - p, n \rangle = 0\},$$

which expands to the familiar equation. Taking $n = (1, 2, 2)$ and $p = (1, 1, 1)$ gives $x + 2y + 2z = 5$, since $\langle p, n \rangle = 1 + 2 + 2 = 5$.

Example 2.15 (From equation to geometry). *The plane $x + 2y + 2z = 0$ passes through the origin and is perpendicular to $n = (1, 2, 2)$. The vector $(2, -2, 1)$ satisfies $2 - 4 + 2 = 0$, so it lies in the plane, consistent with Example 2.11 where we found it perpendicular to n . The distance from a point q to this plane is $|\langle q, n \rangle| / \|n\|$, the length of the projection of q onto the normal; for $q = (3, 0, 0)$ that is $3/3 = 1$.*

2.7 The cross product, briefly

In $\mathbb{R}^3$, and only there, two vectors determine a third perpendicular to both.

Definition 2.16 (Cross product). *For $u, v \in \mathbb{R}^3$,*

$$u \times v = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1).$$

Example 2.17 (A cross product, and two checks). *For $u = (1, 2, 3)$ and $v = (4, 5, 6)$,*

$$u \times v = (2 \cdot 6 - 3 \cdot 5, 3 \cdot 4 - 1 \cdot 6, 1 \cdot 5 - 2 \cdot 4) = (-3, 6, -3).$$

Two sanity checks, both of which you should perform every time: $\langle u, u \times v \rangle = -3 + 12 - 9 = 0$ and $\langle v, u \times v \rangle = -12 + 30 - 18 = 0$. The result is perpendicular to both, as advertised.

The length $\|u \times v\| = \|u\| \|v\| \sin \theta$ is the area of the parallelogram spanned by u and v , which is our first hint that linear algebra measures volume. That hint becomes the determinant of Chapter 8. Note that the cross product is special to three dimensions; the general replacement, valid in every dimension, is the exterior product, and it is Grassmann's idea from the opening of this chapter.

Open Problem 2.18 (How many balls touch a ball?). *Take unit balls in $\mathbb{R}^d$ and place them so that each touches a fixed central unit ball without overlapping any other. The largest possible number of them is the kissing number τ_d . In the plane, $\tau_2 = 6$, and the hexagonal picture is convincing. For most dimensions τ_d is unknown.*

What is known. The exact value has been established in dimensions 1, 2, 3, 4, 8 and 24, and nowhere else. Dimension three, where the answer is 12, was disputed by Newton and Gregory in 1694 and not settled until the twentieth century. Dimensions 8 and 24 fell to a linear programming argument of Levenshtein and Odlyzko and Sloane, using the exceptional lattices E_8 and Leech. Dimension five is open: the answer is known only to lie between 40 and 44 [160].

Why this is a linear algebra problem. The constraint is entirely about inner products. If the touching balls have centres $2v_1, \dots, 2v_\tau$ with each $\|v_i\| = 1$, then no two may overlap, which says $\langle v_i, v_j \rangle \leq \frac{1}{2}$ for $i \neq j$. So the question is: how many unit vectors in $\mathbb{R}^d$ can have all pairwise inner products at most $\frac{1}{2}$? That is a question about the Gram matrix G with $G_{ii} = 1$ and $G_{ij} \leq \frac{1}{2}$, which must be positive semidefinite of rank at most d . The rank constraint is what makes it hard; drop it and the problem becomes a semidefinite program, which is exactly the relaxation that gives the best known bounds.

Where a reader might start. Prove $\tau_2 = 6$ using only the inner product condition: show that seven unit vectors in the plane must have two within 60 degrees. Then try the same argument in $\mathbb{R}^3$ and watch it give 14 rather than 12, which is the whole difficulty in miniature. Chapter 60 explains the semidefinite relaxation that closes most of that gap, and Chapter 53 explains where E_8 and Leech come from.

Exercises

Exercise 2.19. Let $u = (2, -1, 3)$ and $v = (0, 4, -2)$. Compute $u + v$, $u - v$, $3u$, and $\|u\|$.

Exercise 2.20. Decide whether $b = (5, 5)$ lies in $\text{span}\{(1, 2), (3, 1)\}$, and if so find the coefficients.

Exercise 2.21. For which values of t does $\text{span}\{(1, 3), (2, t)\}$ fail to be all of $\mathbb{R}^2$? Justify your answer geometrically as well as algebraically.

Exercise 2.22. Find the angle between $(1, 0, 1)$ and $(0, 1, 1)$. Hint: both have the same length, which simplifies the arithmetic.

Exercise 2.23. Project $b = (2, 3)$ onto the line spanned by $a = (1, -1)$, and verify that the error is perpendicular to a .

Exercise 2.24. Show that equality holds in Cauchy–Schwarz, $|\langle x, y \rangle| = \|x\| \|y\|$, precisely when x and y are parallel, meaning one is a scalar multiple of the other.

Exercise 2.25. Write the equation of the plane through $(2, 0, 1)$ with normal $(3, -1, 2)$, and find the distance from the origin to that plane.

Exercise 2.26. Take the polynomials of degree at most two, with $p + q$ and λp defined coefficientwise as in Example 2.4. Compute $2(1 + t) - 3(t - t^2)$, and decide whether $1 + t^2$ lies in $\text{span} \{1 + t, t + t^2\}$.

Exercise 2.27. Prove that the projection of Definition 2.13 deserves its name: for p the projection of b onto the line spanned by a , show that $\|b - p\| \leq \|b - ta\|$ for every scalar t , with equality only when $ta = p$. Hint: expand $\|b - ta\|^2$ as a quadratic in t .

References and further reading

The historical sketch of Grassmann, Gibbs and Heaviside follows the standard account of how vector analysis reached its modern form; Grassmann’s work was indeed neglected for decades before its significance was recognised. The treatment of span and linear combination here is our own, written to make the reachability question central from the outset; readers wanting a slower, example-first development of the same material will find Hefferon [307] unusually patient, while Austin [35] keeps the geometric picture in view throughout. The dot product, its role in measuring angle, and the projection formula are developed in the context of general inner product spaces in Banica [51], our baseline for the later parts of this book, where they reappear as the foundation of the spectral theory of Part V. Axler [37] treats inner products and orthogonality with unusual care over what depends on finite dimension and what does not.